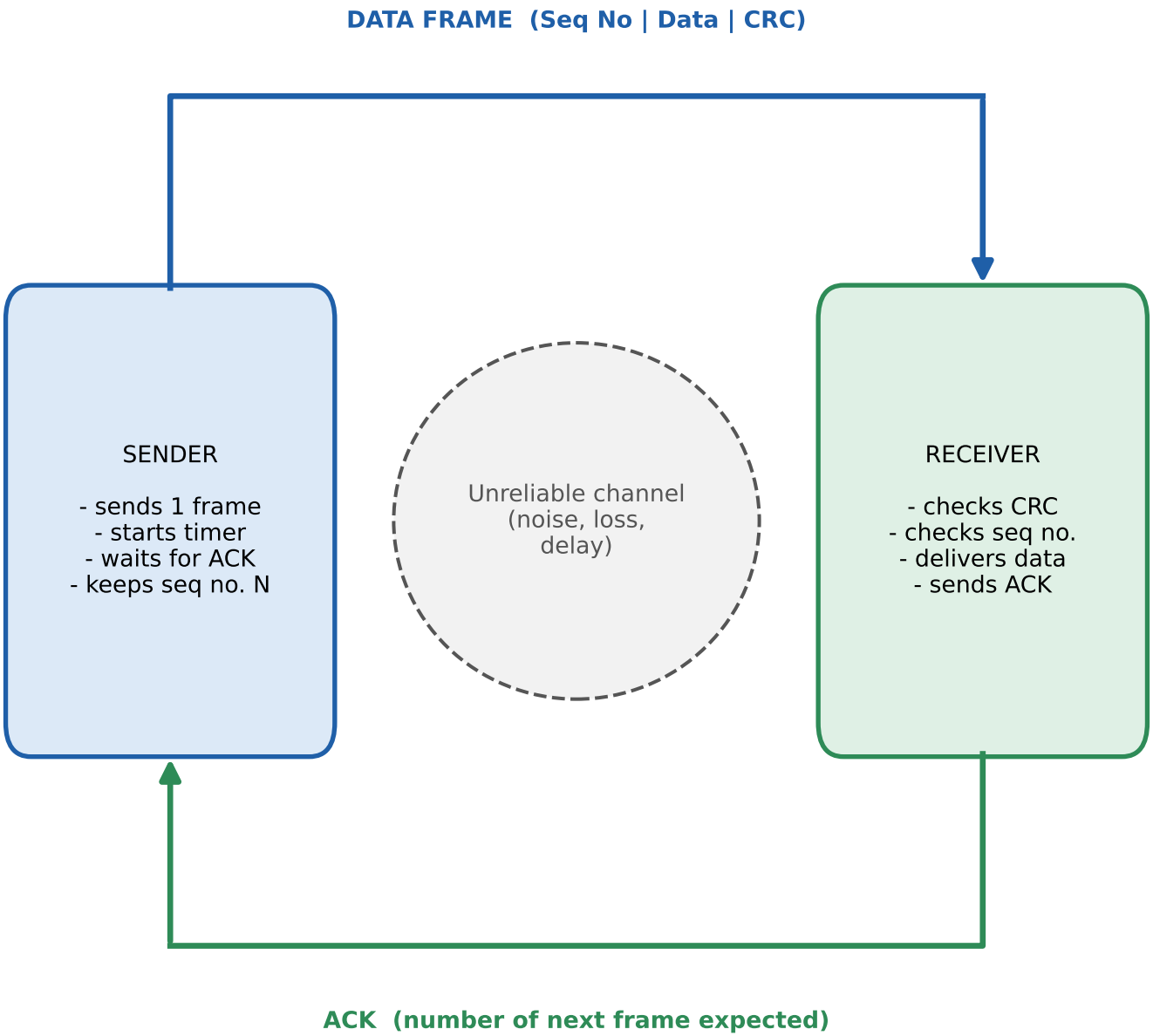
Figure 1: Stop-and-Wait ARQ - System Model

Networking and Web Technology (ESC-AIDS501) | Practical 3: Stop-and-Wait ARQ



Frame format

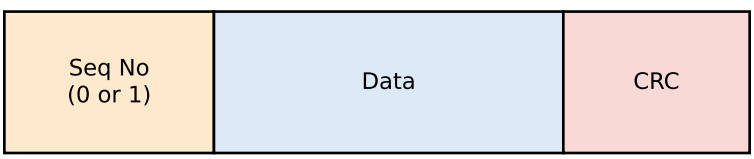


Figure 1. The sender transmits one data frame at a time and starts a timer. The receiver checks the frame (CRC) and replies with an ACK carrying the number of the next frame it expects. The sender sends the next frame only after the ACK arrives; if the timer expires first, it retransmits. Sequence numbers (0 and 1 alternately) let the receiver detect duplicate frames.

Figure 2: Situation 1 - Normal Operation (No Errors)

Networking and Web Technology (ESC-AIDS501) | Practical 3: Stop-and-Wait ARQ

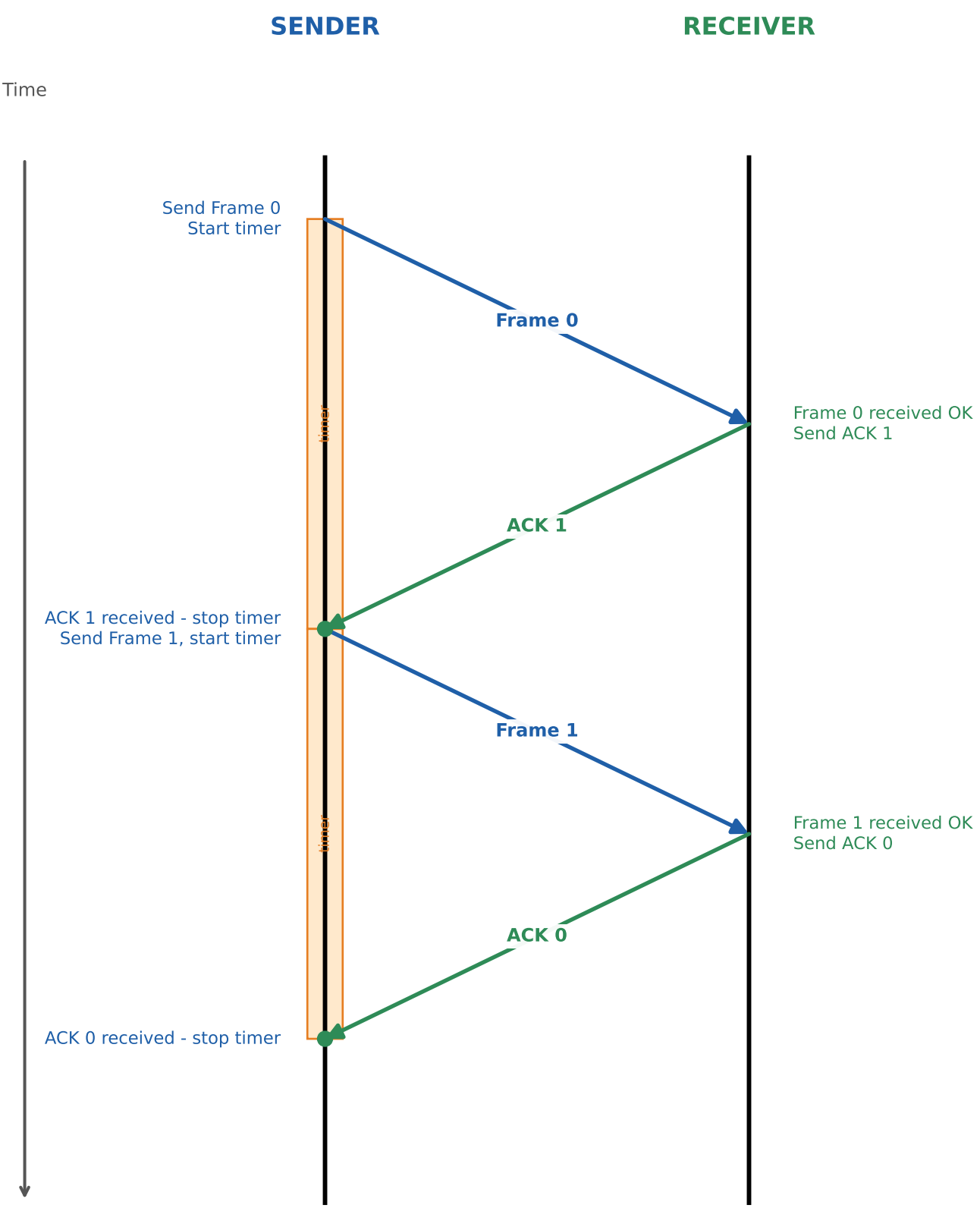


Figure 2. Frame 0 reaches the receiver intact and ACK 1 returns before the timer expires. The timer is stopped (green dot) and the sender moves on to Frame 1. Sequence numbers alternate 0, 1, 0, 1 ... and no retransmission is needed.

Figure 3: Situation 2 - Data Frame Lost or Corrupted

Networking and Web Technology (ESC-AIDS501) | Practical 3: Stop-and-Wait ARQ

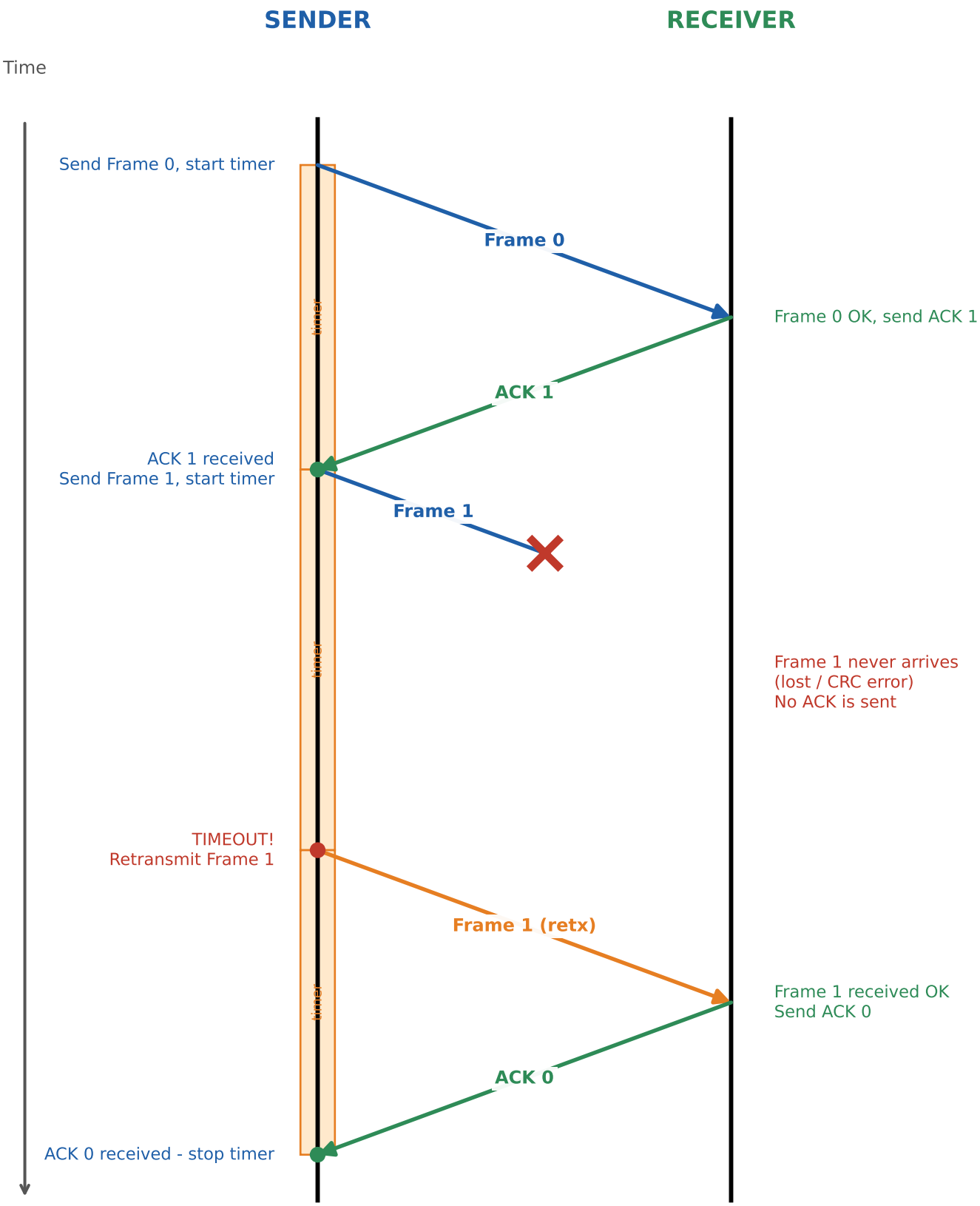


Figure 3. Frame 1 is lost (red cross) or discarded by the receiver after a failed CRC check, so no ACK is generated. The sender's timer expires (red dot), and the sender retransmits Frame 1. The retransmitted frame is accepted and acknowledged normally.

Figure 4: Situation 3 - Acknowledgment (ACK) Lost

Networking and Web Technology (ESC-AIDS501) | Practical 3: Stop-and-Wait ARQ

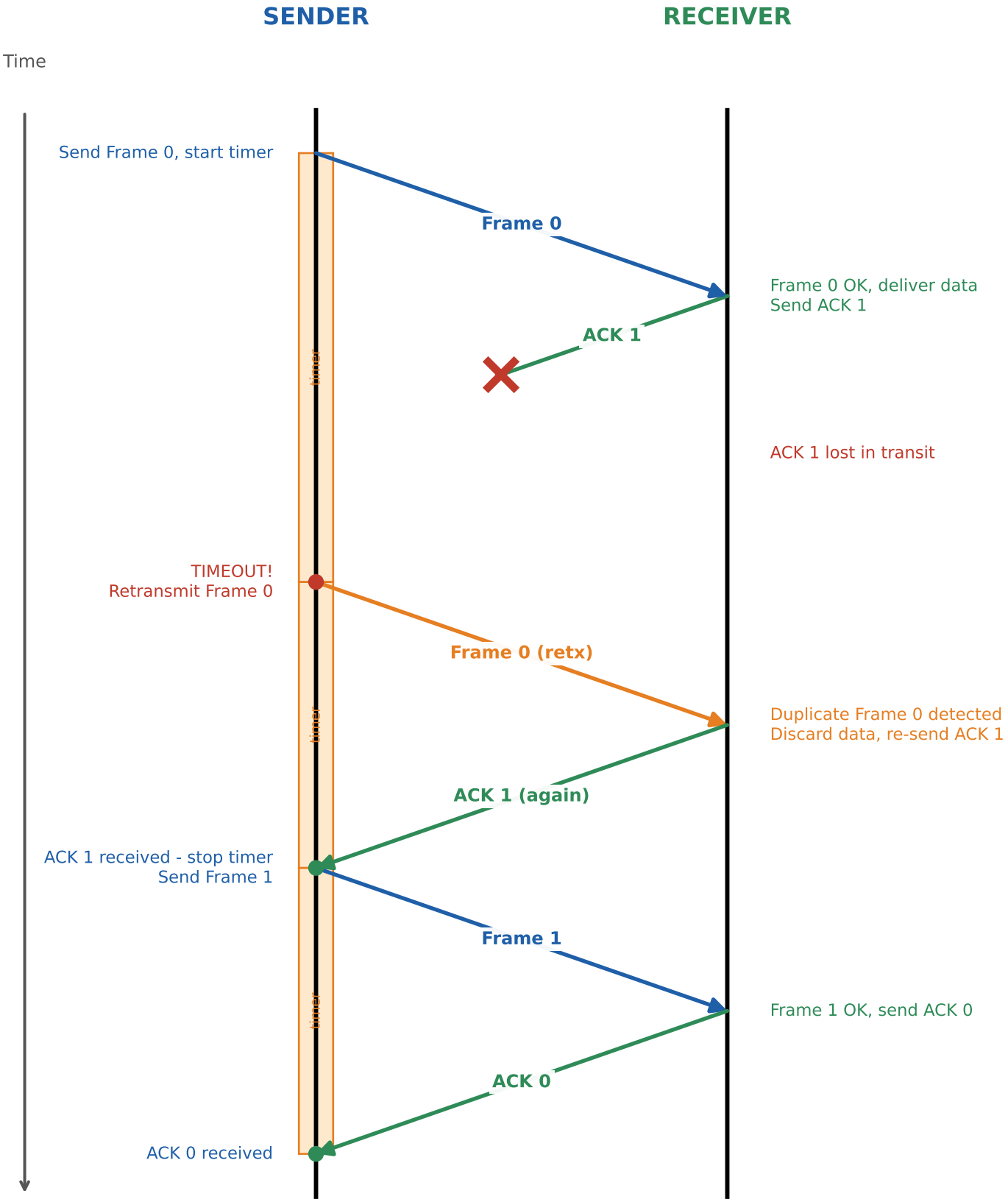


Figure 4. Frame 0 is received correctly but ACK 1 is lost on the return path. The sender cannot tell this apart from a lost frame, so it times out and resends Frame 0. The receiver recognises the duplicate from its sequence number, discards it and re-sends ACK 1.

Figure 5: Situation 4 - Delayed ACK / Premature Timeout

Networking and Web Technology (ESC-AIDS501) | Practical 3: Stop-and-Wait ARQ

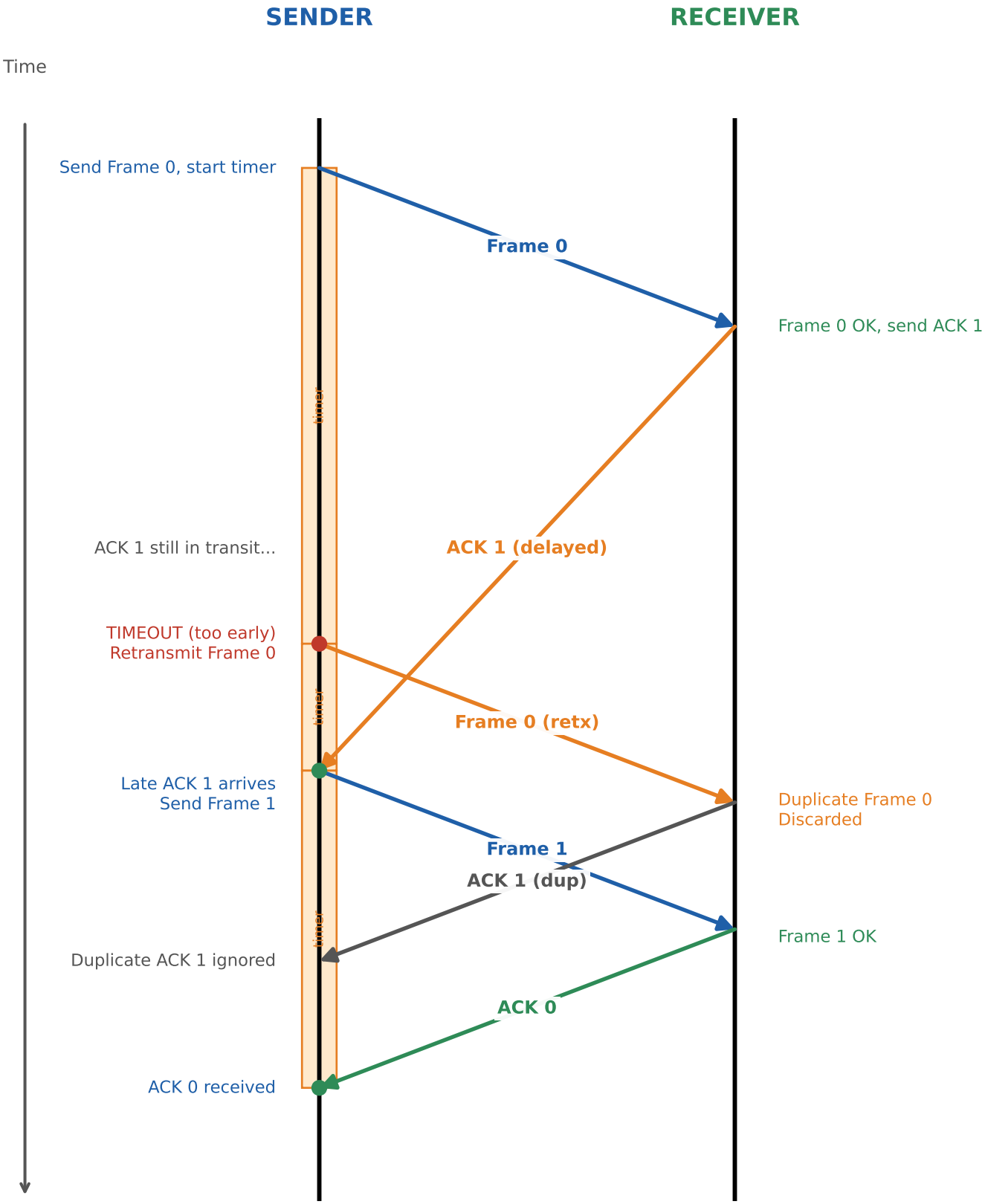


Figure 5. ACK 1 is not lost but is delayed beyond the timeout. The sender has already retransmitted Frame 0, which the receiver discards as a duplicate. The duplicate ACK 1 that comes back is ignored by the sender. This shows why the timeout must be chosen carefully.

Figure 6: Why Stop-and-Wait Is Inefficient - Idle Time

Networking and Web Technology (ESC-AIDS501) | Practical 3: Stop-and-Wait ARQ

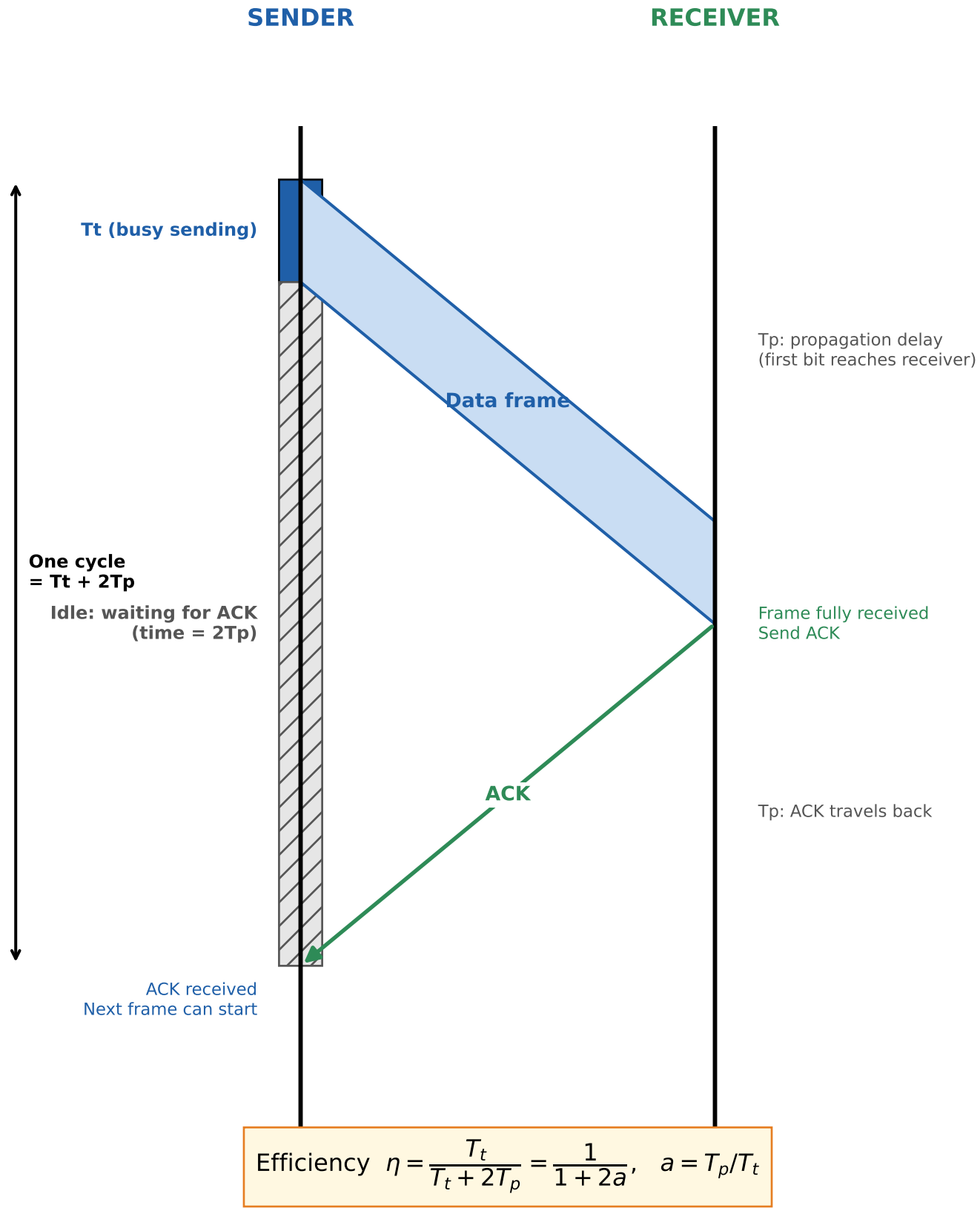


Figure 6. The sender is busy only for the transmission time Tt. For the rest of the cycle (the round trip 2Tp) it waits idle for the ACK. Efficiency = Tt / (Tt + 2Tp) = 1 / (1 + 2a), where a = Tp / Tt. For long links a is large, so efficiency is very low.

Figure 7: Efficiency of Stop-and-Wait vs Link Delay

Networking and Web Technology (ESC-AIDS501) | Practical 3: Stop-and-Wait ARQ

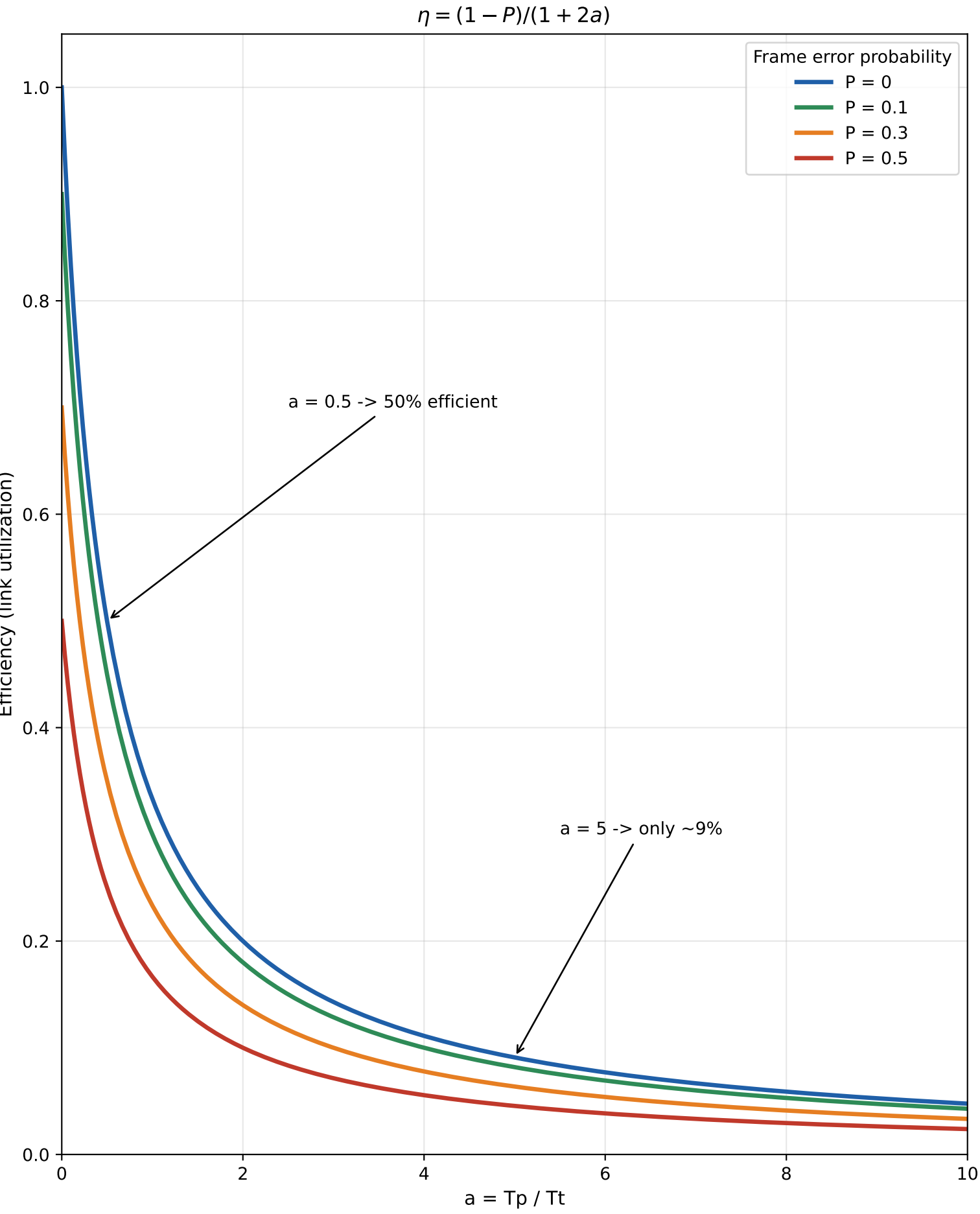


Figure 7. Efficiency falls rapidly as a = Tp/Tt increases (long or fast links). A non-zero frame error probability P reduces it further, since each frame needs on average 1/(1-P) transmissions: efficiency = (1-P)/(1+2a).

Figure 8: Flowchart - Sender Algorithm

Networking and Web Technology (ESC-AIDS501) | Practical 3: Stop-and-Wait ARQ

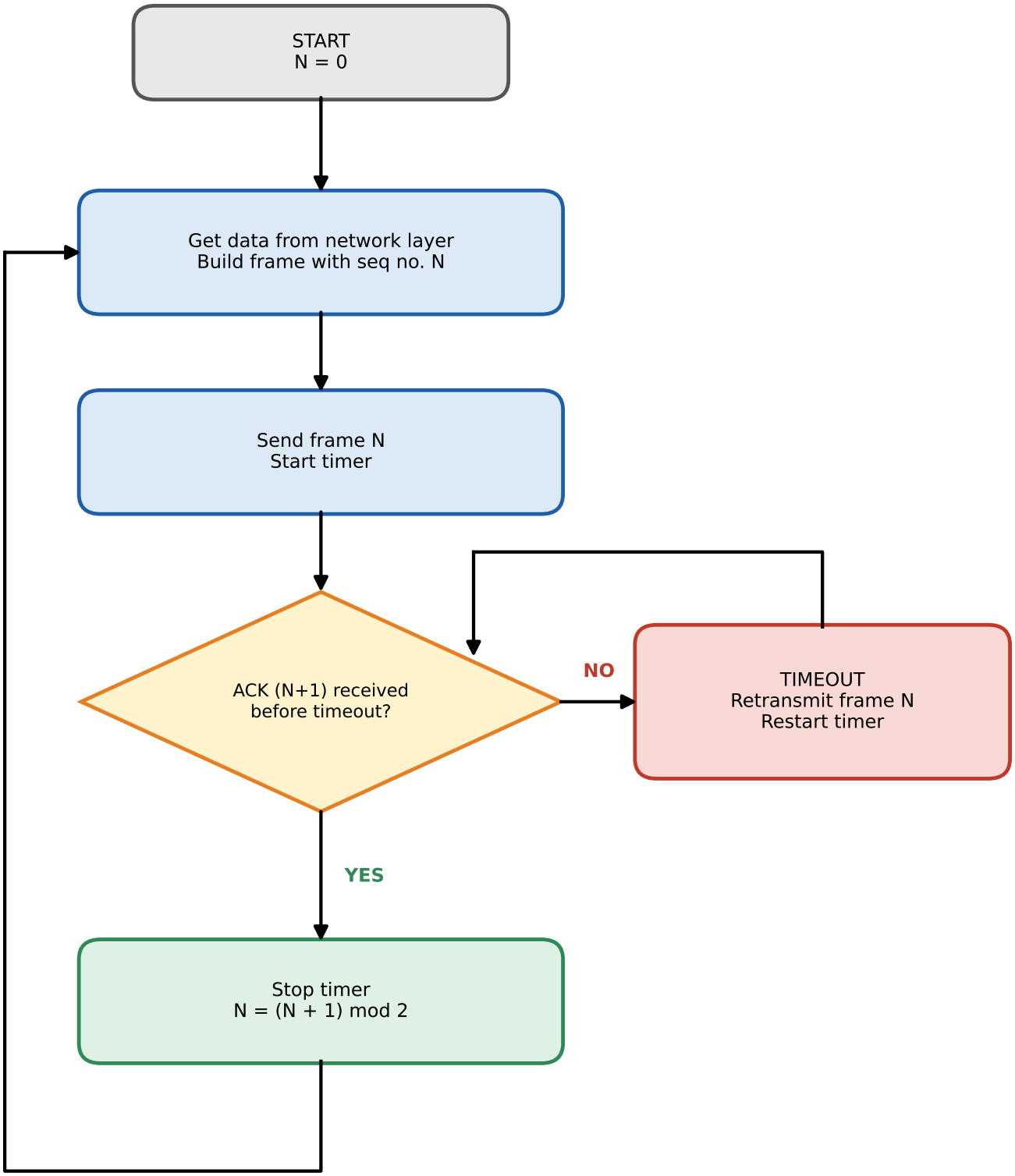


Figure 8. The sender builds a frame with sequence number N, sends it and starts a timer. If ACK (N+1) arrives in time, N is toggled and the next frame is sent; otherwise the timer expires and the same frame is retransmitted.

Figure 9: Flowchart - Receiver Algorithm

Networking and Web Technology (ESC-AIDS501) | Practical 3: Stop-and-Wait ARQ

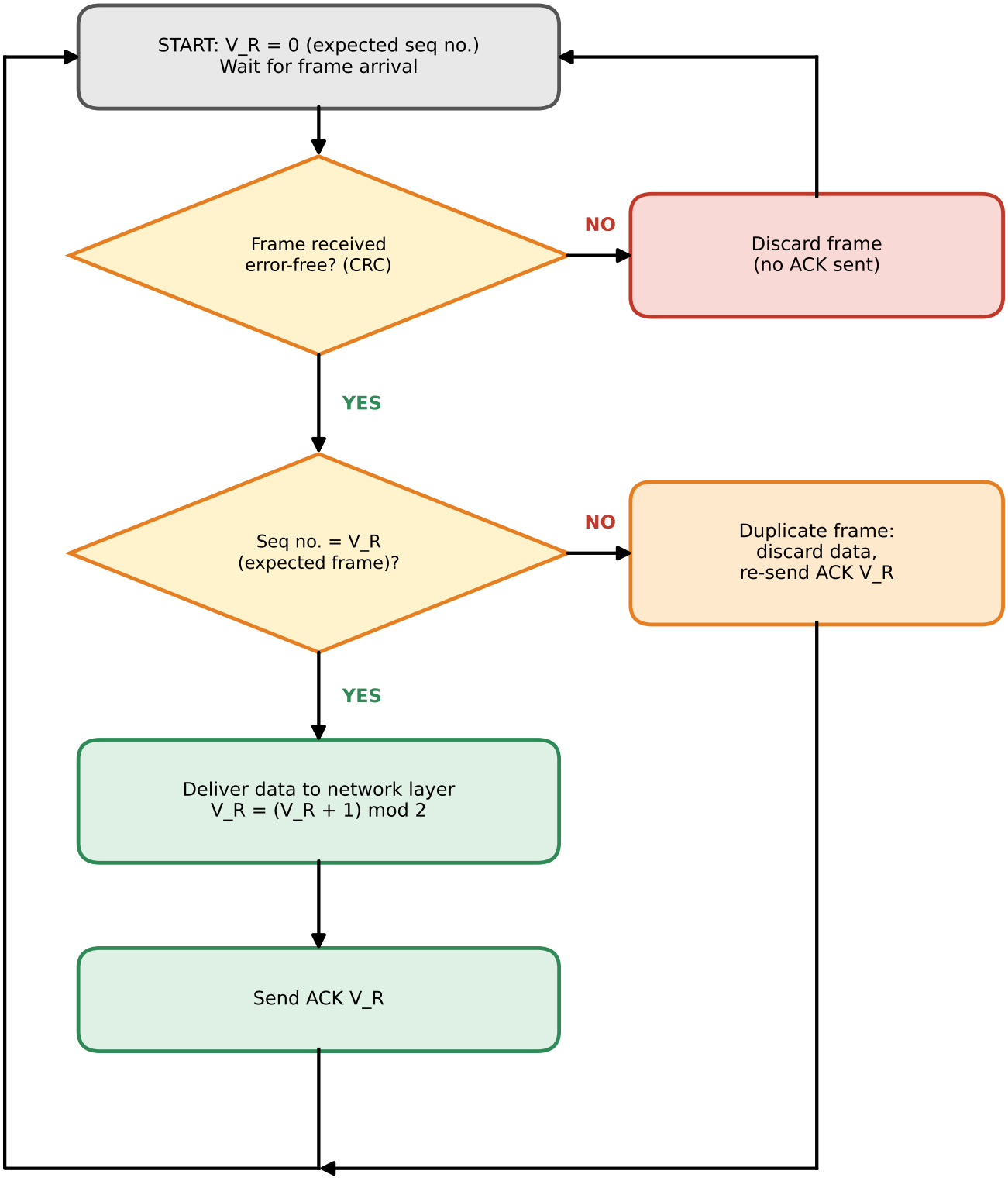


Figure 9. The receiver discards corrupted frames silently (forcing a sender timeout). A correct frame with the expected sequence number is delivered and acknowledged; a correct frame with an old sequence number is a duplicate, so its data is discarded but the ACK is re-sent.